Exercise 8.2

$$f(t) = \begin{cases} 2At & -1/2 \le t < 0 \\ 2A(1-t) & 0 \le t < 3/2 \end{cases}$$

$T_0 = 2$

$W = \pi$

$$b_n = \frac{2}{2} \int_0^{T_0/2} f(t) \sin(n W_0 t)\, dt$$

$$b_n = \underbrace{1 \int_0^{1/2} 2t \sin(n\pi t)\, dt}_{\checkmark} + \int_{1/2}^{1} 2\sin(n\pi t) - 2t \sin(n\pi t)\, dt$$

$u = 2t$

$du = 2dt$

$$= \frac{-2t}{n\pi} \cos(n\pi t) + \frac{2}{n\pi} \int_0^{1/2} \cos(n\pi t)\, dt$$

$v = \sin(n\pi t)\,dt$

$v = -\frac{\cos(n\pi t)}{n\pi}\,d$

$$= \frac{-1}{n\pi} \cos(n\pi/2) + \frac{2}{n^2\pi^2} \sin(n\pi/2)$$

$$\int 2\sin(n\pi t) - 2t \sin(n\pi t)\, dt$$

$$= \left[\frac{-2(1-t)}{n\pi} \cos(n\pi t) + \frac{2}{n\pi} \int_{1/2}^{1} \cos(n\pi t)\, dt \right]_{1/2}^{1}$$

$$= \frac{1}{n\pi} \cos(n\pi/2) - \frac{2}{n^2\pi^2} \sin(n\pi/2)$$

$$b_n = \frac{8}{n^2\pi^2} \sin(n\pi/2)$$

$a_n = 0$

$a_0 = 0$

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8.3:

Function is odd

$a_0 = 0$
$a_n = 0$

$$f(t) = \begin{cases} \frac{2t}{\pi}, & -\pi/2 \leq t \, 0 \\ \frac{-2t}{\pi} + 2 & 0 \leq t \leq \pi/2 \end{cases}$$

$T_0 = 2\pi$

$W_0 = 1$

$$b_n = \frac{1}{\pi}\int_{-\pi/2}^{\cancel{\pi/2}\;0} \frac{2t}{\pi}\sin(nt) + \frac{1}{\pi}\int_0^{\pi/2}\left(-\frac{2t}{\pi}+2\right)\sin(nt)\,dt$$

$$= \frac{1}{\pi}\int_0^{\pi/2} \frac{-2t}{\pi}\sin(nt)\,dt + \int_0^{\pi/2} 2\sin(nt)\,dt$$

$u = t$

$du = 1dt$

$dv = \sin(nt)$

$v = \int -\frac{\cos(nt)}{n}$

$$\frac{2}{\pi^2}\left[\frac{-t\cos(nt)}{n} - \int -\frac{\cos(nt)}{n}\,dt\right] + \frac{2}{\pi}\left[-\frac{\cos(nt)}{n}\right]_0^{\pi/2}$$

$$= \frac{2}{\pi^2}\left(\frac{-t\cos(nt)}{n} + \frac{\sin(nt)}{n^2}\right] + \frac{2}{\pi}\left[-\frac{1}{n} + \frac{1}{n}\right]$$

$$= \frac{2}{\pi^2}\left[\frac{-t\cos(n\pi/2)}{n} + \frac{\sin(n\pi/2)}{n^2}\right]$$

$$= -\frac{2\pi/2\,\cancel{\cos}(1)}{\pi^2 n}$$

$$= \frac{2}{2n} = \frac{1}{n}$$

$$\int_{-\pi/2}^{0} \frac{2t}{\pi} \sin(nt)\,dt$$

$u = t$

$du = 1dt$

$dv = \sin(nt)$

$v = -\frac{\cos(nt)}{n}$

$$= \frac{2}{\pi}\left[-\frac{t\cos(nt)}{n} - \int \frac{-\cos(nt)dt}{n}\right]$$

$$= \frac{2}{\pi}\left[-\frac{t\cos(nt)}{n} + \frac{\sin(nt)}{n^2}\right]_{-\pi/2}^{0}$$

$$= \frac{2}{\pi}\left[\frac{+\pi/2\cos(n\pi/2)}{n} \cancel{+\sin(}\right.$$

$$= \frac{2}{\pi}\left[\left(\frac{0\cos(0)}{n} + \frac{\sin(0)}{n^2}\right) - \left(\frac{+\pi/2\cos(-n\pi/2)}{n} + 0\right)\right]$$

$$= \frac{2}{\pi}\left[-\frac{\pi/2\cos(n\pi/2)}{n}\right]$$

$$= -\frac{2\pi(1)}{2\pi n} = \frac{-2}{n}$$